

The role of word order in the diachrony of flagging

Martin Haspelmath

revised, June 2026

This article challenges the widespread idea that case affixes generally evolve from adpositions through increasing “bondedness.” Instead, I propose that word order, not only increasing tightness of attachment, is another primary factor determining whether a marker becomes an affix or remains a clitic. Using a simple and clear definition of *clitic*, the paper notes that flags are affixes (case markers) when there is no adnominal modifier occurring between them and the noun, but adpositions otherwise. In view of this, it may be that we need to revise our thinking about grammaticalization, because what all grammaticalization changes share is the development of bound markers from free contentful forms, not necessarily notions like coalescence, fusion or bondedness.

1. From adposition to case affix: Increasing bondedness?

This paper asks whether case affixes typically derive from adpositions by a process of increasing bondedness (or increasing “coalescence” or “fusion”). It is generally thought that the type of change schematized in (1) is a widespread development in languages that come to have case affixes, and that this is a typical grammaticalization change.

- (1) adposition > case affix
(clitic flag > affix flag)

According to the bondedness view, this kind of change is an instance of the general “cline of grammaticality” schema in (2), which involves “a unidirectional progression in bondedness, i.e. in the degree of cohesion of adjacent forms ranging from loosest (“periphrasis”) to tightest (“morphology”)” (Hopper & Traugott 2003: 7).

- (2) content item > grammatical word > clitic > inflectional affix

While “bondedness” is a term that goes back to Lehmann (2015: §4.3.2) (originally published in 1982) and has not been universally adopted,¹ the idea that clitics are typically intermediate between lexemes and affixes is very widespread. For example, Spencer & Luís (2012: §2.1) say that “inflections are generally thought to arise in historical language change through a stage of cliticization”, and a cline as in (2) is widely assumed in the literature (e.g. Wiemer 2014: 442; Konvička 2019; Narrog & Heine 2021: 279; Bisang et al. 2020: 37).

In this paper, I question the generality of the diachronic pathway in (2), as well as the importance of increasing bondedness (these questions were first asked in a blogpost, Haspelmath 2023b). Instead, I highlight the importance of word order for the distinction between clitics and affixes, which has not been discussed widely before. The two key observations on which the remainder of this paper is based are summarized in (3a-b).

¹ Some related terms are “cohesion”, “fusion”, “morphophonological integration”, “tightness of attachment”, “coalescence”, “agglutination”. These terms have somewhat different connotations, but they have not been made precise, and the main reason for using *bondedness* here is that it has been used in the more rigorous typologically oriented approaches (Lehmann 2015; Bisang et al. 2020: 37).

- (3) a. The difference between affixes and clitics cannot generally be characterized in terms of “tightness of bonding”, as has sometimes been suggested.
- b. Affixes are defined as class-selective, so for flags, adjacency to the noun is crucial, and this depends on the order of modifiers.

In the earlier literature, the terms “affix”, “clitic” and “adposition” are often characterized only vaguely. In this paper, I use the definitions of these terms from my earlier work (Haspelmath 2021; 2023a). Affixes and clitics are defined as in (4a-b), and adpositions are defined as in (5).

- (4) a. **affix** (Haspelmath 2021: 8)
An affix is a bound morph that is not a root, that must occur on a root, and that cannot occur on roots of different root classes.
- b. **clitic** (Haspelmath 2023a: 2)
A clitic is a bound morph that is neither an affix nor a root.

- (5) **adposition**
An adposition is a clitic flag.

A flag is defined as “a grammatical marker that occurs on a nominal and that indicates its semantic role or syntactic function” (Haspelmath 2027a). These definitions conform largely to linguists’ practice in the literature, though the term “clitic” has often been associated with a notion of “prosodic dependence” that plays no role in the definition here (see §2 below). And adpositions have often been characterized more loosely, as including also free (non-clitic) forms such as English *in front (of)* or French *derrière* ‘behind’. Such free forms for relational meanings are disregarded here, and it is assumed that it is uncontroversial that the shared core of the term “adposition” comprises bound markers such as *in*, *at*, *on*, *from*, or French *dans*, *à*, *sur*, *de*.

To illustrate the importance of the order of nominal modifiers, let us look briefly at Turkish and Basque. Turkish has prenominal modifiers such as adjectives, and flags (e.g. *-de* ‘in’) always follow the noun, so they are suffixes. In Basque, by contrast, adjectives follow the noun, so flags like *etan* ‘in’ are not class-selective (they can follow nouns or adjectives), and hence they are not suffixes but clitics (i.e. postpositions).²

- (6) Turkish: (adjective +) NOUN + **flag**
büyük ev-ler+de (case suffix)
 big house-PL+in
 ‘in big houses’
- (7) Basque: NOUN (+ adjective) + **flag**
etxe handi + etan (postposition)
 house big + in.PL
 ‘in the big houses’

² Some postpositions are written as if they were suffixes in Basque, but their occurrence at the end of a nominal (= a nominal phrase) makes them clear cases of clitics.

We will see more examples of word order differences as crucial for the difference between clitic flags and affix flags below. Of course, the role of word order that I emphasize here is not necessarily incompatible with the traditional view of increased bondedness, but the purpose of this article is to highlight the role of word order and word order change, and to shift the burden of proof to the bondedness view.

It should be clear by now that the difference between (what I call) the “bondedness view” in (2) and the word-order perspective emphasized in this paper is partly a matter of definition. Here I use the definitions in (4) and (5), and if someone does not want to use the terms “affix” and “clitic” in this way, they will see quite a few things differently. However, these definitions come fairly close to the consensus in the literature, even though affixes and clitics are stereotypically also associated with phonological properties that play no role in the definitions. In any event, these different perspectives should ultimately be reconciled, and hopefully the present paper provides some food for thought about ways in which they may be brought together.

I should also note that the importance of word order in grammaticalization and wordhood has not gone unnoticed in the past. Himmelmann (2014: 932) notes that one requirement for affixalization is “exclusive adjacency”:

“In order for function words to become affixes it is necessary that function words and their hosts regularly (and frequently) occur immediately adjacent to each other. This condition is not met when other elements can intervene between function word and host.” (Himmelmann 2014: 932)

However, Himmelmann also says that “mere positional adjacency is not enough: There also has to be prosodic integration for further coalescence to occur”, but what exactly “prosodic integration” means in general remains vague. Himmelmann does not envisage the possibility of word order changes leading to changes in affix vs. clitic status.

2. Does phonological shortness lead to prosodic dependence and affixhood?

There is a widespread view according to which clitics are less “intimately attached” or less “tightly bonded” than affixes, which may be indicated by linking them with an equals sign ($H=cl$) to their host rather than with a hyphen ($H-sf$). But “tight attachment” is a vague notion, and “bondedness” and “fusion” are not always clearly defined either.³ The process of change from clitic to affix is widely thought to involve phonological reduction and increased prosodic dependence, but what exactly is meant by these terms is not very clear either. In earlier work (Haspelmath 2011b), I used the term “coalescence”, but I did not define it precisely, and I would not use it again.⁴

While it is often said that clitics are “prosodically dependent”, obligatorily occurring within the domain of an adjacent independent element, this is not true of all clitics. In fact, many clitics are perfectly well-formed prosodically, e.g.

³ See also Haspelmath (2022) for some discussion of the term “fusion” as used in the earlier literature.

⁴ A reviewer says that “simple clitics (English auxiliaries and pronouns being prototypical examples) often involve coalescence without requiring exclusive adjacency. The real challenge is to be able to account for why some clitics coalesce and others do not”, but the reviewer does not offer a clear definition of “coalescence” either. English “simple clitics” (such as *’s* in *Joan’s here*) are shortened variants of longer forms which are clitics as well (see Haspelmath (2023a: §4.3) on “shortened clitics”). It is not clear in what sense they involve “coalescence” rather than merely shortening.

- (8) English *he* (weak pronoun)
 German *halt* (discourse particle)
 Russian *li* (question particle; see (11) below)

These elements are clitics in that they cannot occur on their own, but this does not seem to be directly related to their segmental makeup.

A clearer concept is phonological shortening, and we do indeed repeatedly observe this in diachronic changes involving flags, e.g.

- (9) Latin *dē* ‘from’ > French *də* ‘of’ (*de*)
 Old English *tō* ‘to’ > English *tə* ‘to’ (*to*)
 Ancient Greek *eis* ‘to’ > M. Greek *s* ‘to’

However, these prepositions have not become case prefixes, so their phonological reduction did not cause a change from clitic to affix. They are still prepositions, as they occur not only next to nouns, but also next to prenominal modifiers (see (11)-(13) below).

While prosodic dependence is often mentioned in definitions of “clitic”, it has not often been discussed in the context of the assumed grammaticalization path from clitic to affix. An exception is Himmelmann (2014), perhaps the best-known work that links prosodic features to affixalization. The author says:

“[I]t is shown that preposed function words are more often and more systematically separated from their lexical hosts by a PROSODIC BOUNDARY. ... The prosodic boundary, according to the hypothesis, is a major factor in BLOCKING THE FUSION of a preposed function word and its lexical host.” (Himmelmann 2014: 955; emphasis added)

Himmelmann assumes that affixalization involves phonological “fusion” of some kind. However, adpositions are typically already integrated phonologically with their complement nominal. Thus, they have no separate stress and are typically unstressed, and they often show sandhi effects (for example, in English *for us* and in French *dans elle* ‘in her’, the final consonant letter is pronounced). In quite a few languages, prepositions are sometimes subsyllabic, e.g. French *d* ‘of’, Hebrew *b-* ‘in’, Polish *z* ‘with’, Maltese *f-* ‘in’.

Some adpositions even show lexically conditional phonological alternations, e.g. Russian prepositions such as *s* ‘with’, *k* ‘to’, *v* ‘in’ (e.g. Linzen et al. 2013). The shape of these prepositions depends not only on the phonological environment (as in the first two lines of (10)), but also on the specific words they combine with (as in the third line).

- (10) *v nužde* ‘in need’ *vo vrede* ‘in harm’ (**v vrede*)
s radost’ju ‘with joy’ *so straxom* ‘with fear’ (**s straxom*)
s mneniem ‘with opinion’ *so mnoj* ‘with me’ (**s mnoj*)

To conclude, it appears that phonological properties of short elements cannot easily explain why they become affixes. Increased frequency of use and grammaticalization may lead to segmental reduction and shortness, but it does not automatically lead to affixalization. We saw the examples of French *de*, English *to* and Greek *s* above in (9). Why haven’t these prepositions become prefixes? This does not seem to have prosodic reasons, but the explanation is simply that they do not always occur on nouns. These three languages have prenominal modifiers, and we find examples such as (11b), (12b) and

(13b), in which the prepositions do not occur on the noun directly. Thus, these flags are prepositional clitics, not prefixes.

(11) French

- a. *de Gilles*
'of Gilles'
- b. *de la nouvelle maison*
'of the new house'

(12) English

- a. *to Paris*
- b. *to the interesting workshop*

(13) Modern Greek

- a. σ' *αυτόν*
s aftón
'to him'
- b. σ' *ένα παιδί*
s ena peði
'to one child'

The next section elaborates on the distinction between clitics and affixes on the basis of selectivity.

3. Affixes are class-selective, clitics are not

Since Zwicky & Pullum (1983), it has been widely accepted that affixes are distinguished by exhibiting host-class selectivity (summarized in (14)): They can occur only on nouns, or only on verbs, or only on adjectives, while clitics are "promiscuous" – they can occur on roots of different classes.

(14) Affixes are class-selective, clitics are nonselective.

The typical behaviour of clitics can be illustrated by the English genitive clitic 's and the Russian polar question particle *li* (as used in formal language):

(15) English

- a. *my friend's house* (adjacent to noun)
- b. *the lady I met yesterday's offer* (adjacent to adverb)
- c. *the boy I like's new bike* (adjacent to verb)

(16) Russian

- a. *Прочитала ли Анна книгу?*
Pročita-la li Anna knigu? (adjacent to verb)
read-PST PQ Anna book
'Did Anna read a book?'

- b. *Книгу ли Анна прочитала?*
Knigu li Anna pročita-la? (adjacent to noun)
 book PQ Anna read-PST
 ‘Did Anna read a BOOK?’
- c. *Вчера ли Анна читала?*
Včera li Anna čita-la? (adjacent to adverb)
 yesterday PQ Anna read-PST
 ‘Did Anna read YESTERDAY?’

Clitics are not necessarily “less bonded” than affixes, and in fact, the English genitive clitic *’s* does not differ phonologically from affixes with the shape *-s*.

In this paper, the overall focus is on flags, but there are also other kinds of grammatical markers that are sometimes clitics and sometimes affixes, depending on word order. For example, definite articles can be suffixes, as in Swedish (in 17), or clitics, as in English (in 18). The difference is that Swedish definite articles always occur next to the noun, and adjectives occur on the other side of the noun. (Here and in the following, I use a plus (+) symbol for linking a bound grammatical marker to its adjacent form; when there is no space around +, the marker is an affix.)

- (17) Swedish
 a. *kopp+en* ‘the cup’
 b. *den nya kopp+en* ‘the new cup’

- (18) English
 a. *the + cup*
 b. *the + new cup*

Plural markers can be suffixes as in Hungarian (in 19), or enclitics, as in Haitian Creole (in 20). The difference is that Hungarian adjectives precede the noun and plural markers always occur after it, while Haitian Creole adjectives follow the noun.

- (19) Hungarian
 a. *fá+k* ‘trees’
 b. *nagy fá+k* ‘big trees’
- (20) Haitian Creole (Fattier 2013)
 a. *liv + yo* ‘books’
 b. *manchèt plat + yo* ‘straight machetes’

Aspect markers can be prefixes, as in Papiá Kristang (progressive prefix *ta-* in (21); see Baxter 1988: 128-129), or proclitics, as in English (in 22). The difference is that Papiá Kristang adverbs follow the verb and the aspect marker immediately precedes the verb, while the English aspect marker does not always occur on the verb because adverbs (such as *now*) may precede the verb.

- (21) Papiá Kristang
 a. *eli ta+les buku* ‘she is reading books’
 b. *eli ta+les buku agora* ‘she is reading books now’

(22) English

- a. *she is + reading a book*
 b. *she is + now reading a book*

Relativizers can be suffixes (forming participles), as in Turkish (in 23), or enclitics, as in Mandarin (in 24). Objects always precede the verb in Turkish, so that the relativizer follows the verb immediately, while relativizers follow the verb or the verb plus its object in Mandarin.

(23) Turkish

pencere-yi aç+a+n kadın
 window-ACC open+REL woman
 ‘the woman who opened the window’

(24) Mandarin Chinese

打开窗户的女人
dǎkāi chuānghù + de nǚrén
 open window REL woman
 ‘the woman who opened the window’

Finally, complementizers can be affixes (forming nominalizations), as in Basque (in 25), or clitics, as in Indonesian (in 26). Complementizers always follow the verb directly in Basque, while Indonesian complementizers can occur next to a diverse set of word types.

(25) Basque (Hualde & Ortiz de Urbina 2003: 635)

etxe-ra doa+la
 house-ALL go+COMP
 ‘that she is going home’

(26) Indonesian (Sneddon 1996: 292)

bahwa + dia akan pulang
 COMP she PROG go.home
 ‘that she is going home’

These pairs of languages and examples illustrate the affix vs. clitic nature of a range of grammatical marker types. It has been seen that the main definitional property of affixes is that they always occur on words of the same class (always on nouns or always on verbs), while clitics are not class-selective. The phonological properties of affixes and clitics do not seem to play a clearly discernible role.

Thus, word order has an important role in determining whether a grammatical marker is an affix or a clitic. This means that word order change can potentially lead to change from clitic to affix, and an instance of this can be seen in the development of object person forms in Spanish. Compare the examples in (27) and (28) (from Pescarini 2021: §1.4.1)

(27) Old Spanish

sy el físico la bien connoſçe
 if the physician 3.F.ACC well know. 3SG
 ‘if the physician knows it well’

(clitic, non-selective)

(28) Modern Spanish

<i>si</i>	<i>el</i>	<i>médico</i>	<i>la-conoce</i>	<i>bien</i>	
if	the	physician	3.F.ACC-know.3SG	well	
'if the physician knows it well'					(affix, selective)

In Modern Spanish, the order of the adverb changed, from preverbal to postverbal, and as a result, the object clitic has become an object prefix. Again, the change had nothing to do with the nature of the pronoun forms themselves.

Before we move on, I should mention that the literature on clitics does not typically talk about all of the elements that are treated as clitics here, so some of what I have said here may seem surprising. For example, complementizers and definite articles are not often discussed under the “clitic” heading. However, if we want to have a coherent definition of the term *clitic*, we can hardly leave them out.⁵ The most striking clitics are of course those that occur in an unexpected position (second-position clitics, such as Russian *li* in (16)), but postposed plural markers such as Haitian *yo* in (20) and postpositions such as English genitive *'s* are also generally regarded as clitics. I am assuming that any coherent definition of the term *clitic* will have to include definite articles, coordinators, complementizers, and adpositions as well.

4. Flags are suffixes when the noun must occur in final position

In some languages, nominal modifiers always precede the noun, which means that postposed flags always occur on the noun, i.e. they are suffixes. The “degree of bondedness” plays no separate role in these languages. One example is Lezgian, a language described in Haspelmath (1993) and Gajdarov et al. (2009). The examples below illustrate prenominal numerals (29a), demonstratives (29b), adjectives (29c), adpossessionors (29d), and relative clauses (29e). There are no postnominal modifiers at all in Lezgian, so flags (such as *-a* in (29d)) always occur immediately adjacent to the noun.

(29) Lezgian (Haspelmath 1993: 251)

a. *Za-z q'we nük' aku-na.*
 I-DAT two bird see-AOR
 ‘I saw two birds.’

b. *a insan-ar*
 that human-PL
 ‘those people’

c. *ask'an k'waler*
 low house
 ‘a low house’

⁵ One reviewer thinks that the present paper is “anachronistic” because the literature on clitics has shown that clitics are not a coherent phenomenon (especially Spencer & Luís 2012; see also Haspelmath 2015). However, on the definition that I use here (Haspelmath 2023a), clitics are coherent by definition, which allows me to evaluate claims such as those in (2) (the “cline of grammaticality”) which are widely believed to be true. If “clitic” were not a coherent notion, then the claim in (2) would not even be coherent.

- d. *Dağustan.di-n šeher-r-a*
 Daghestan-GEN city-PL-INCESS
 ‘in the cities of Daghestan’
- e. *wa-l tapšurmiš-da-j sa c’iji kar*
 [you-SRESS entrust-FUT-PTCP] one new job
 ‘a new job that will be entrusted to you’

Because of this word order regularity, all flags must be suffixes – not only those that have traditionally been called “case suffixes” (some of which are illustrated in 30a; Haspelmath 1993: 74), but also forms that have traditionally been called “postpositions” (some of which are illustrated in 30b).

- (30) a. Absolutive *sew* ‘the bear’
 Ergative *sew-re* ‘the bear’
 Genitive *sew-re-n* ‘of the bear’
 Dative *sew-re-z* ‘to the bear’
 Adessive *sew-re-w* ‘at the bear’
 Subessive *sew-re-k-aj* ‘from under the bear’
- b. *patal:* *sew patal* ‘for the bear’
q’iliw: *sew-re-n q’iliw* ‘to near the bear’
pataj: *sew-re-n pataj* ‘from the bear’

The flags that have been called “postpositions” are written separately, and they tend to be different from the “case suffix” flags in that they are longer (bisyllabic instead of monosyllabic or nonsyllabic), they may look internally complex (*pat-al*, *pat-aj*, *q’il-iw*), and they may combine with a form of the noun that already contains a case suffix. However, none of these properties can be used to differentiate adpositions from case affixes in general: Affixes may of course be bisyllabic, they may be internally complex (as also illustrated by the Subessive *-k-aj* in (30a)), and they may combine with a special form of the noun (as also illustrated by Lezgian case suffixes such as *-n* and *-z*, which combine with a special “oblique” form coinciding with the Ergative case).

It may seem surprising to see Lezgian described in this way, but for a better-known language, Hungarian, a very similar view has been repeatedly articulated. As noted by authors such as Creissels (2006), Spencer (2008), Trommer (2008), and Thuilier (2011), the “case suffixes” and “postpositions” of Hungarian sometimes behave in unexpected ways, and they are not clearly different from each other. For example, not only case suffixes, but also postpositions are repeated on prenominal demonstratives, e.g. *-ben/-ban* (locative) in (31a) and *mögött* (‘behind’) in (31b).

- (31) Hungarian
- a. *e-bben a ház-ban*
 this-LOC the house-LOC
 ‘in this house’ (lit. ‘in-this the in-house’ also from Spencer 2008?)
- b. *e mögött a ház mögött*
 this behind the house behind
 ‘behind this house’ (lit. ‘behind-this the behind-house’; Spencer 2008: 40)

Creissels (2006) and Spencer (2008) (as well as others) suggested that Hungarian case suffixes such as locative *-ban* are really postpositions, but alternatively one should probably say that they are not postpositions but suffixes, because they all occur directly after a noun.⁶ As in Lezgian, adjectives, numerals and other nominal modifiers precede the noun in Hungarian.

- (32) a. *az új törvény-hez*
 the new law-ALL
 ‘to the new law’
- b. *a három törvény-hez*
 the three law-ALL
 ‘to the three laws’

A modifier such as *óvodák-ról* [kindergarten-PL-DELTATIVE] ‘about kindergartens’ cannot follow the noun and precede the “case/postposition”, as can be seen in (33a). Instead, modifiers must precede it (together with a participial form) as in (33b), or follow the noun + case/postposition as in (33c).

- (33) a. **az új törvény óvodák-ról hez*
 the new law kindergartens-about ALL
 ‘to the new law about kindergartens’
- b. *az óvodák-ról szóló új törvény-hez*
 the kindergartens-about concerning new law-ALL
 ‘to the new law about kindergartens’
- c. *az új törvény-hez, ami-t óvodák-ról ír-t-ak.*
 the new law-ALL which-ACC kindergartens-about write- PST-3PL
 ‘to the new law that was written about kindergartens’

Thus, the postposed flags always occur on nouns, supporting the idea that they are suffixes, like the Lezgian flags. This applies both to traditional suffixes such as *-hez* ‘to’ and to traditional postpositions such as *mögött* ‘behind’. (For full details on Hungarian postpositions, see É. Kiss & Hegedűs (eds) (2021).)

Hungarian can be compared with Northern Saami, where not all nominal modifiers are preposed (Julien 2018). A nominal phrase such as ‘the new law for kindergartens’ can have a postposed postpositional modifier, as seen in (34c). The corresponding Hungarian expression would not be possible.

- (34) Northern Saami (Julien 2018: §2-3)
- a. *ođđa lága birra*
 new law about
 ‘about the new law’

⁶ And as noted by Thuilier (2011), the phonological behaviour of the Hungarian flags does not correlate clearly with the traditional distinction between case suffixes and postpositions that is suggested by the spelling.

- b. *mánáidgárddiid* *várás*
 kindergartens.GEN.PL for
 ‘for kindergartens’
- c. *ođđa lága mánáidgárddiid* *várás*
 new law [kindergartens.GEN.PL for]
 ‘the new law for kindergartens’

However, when a nominal phrase that contains a postposed positional modifier is followed by a postposition, the result is ungrammatical in Saami, as seen in (35a). Two postpositions cannot follow each other. Preposing the modifier instead of postposing it is not possible either, as seen in (35b).

- (35) a. no postposition following postnominal modifiers:
 **ođđa lága mánáidgárddiid* *várás birra*
 new law [kindergartens.GEN.PL for] about
 ‘about the new law for kindergartens’
- b. no preposed PP modifiers:
 **mánáidgárddiid* *várás ođđa lága birra*
 [kindergartens.GEN.PL for] new law about
 ‘about the for kindergartens new law’

The only possibility, according to Julien (2018: 160), is to postpose the entire postpositional modifier after the postposition, as seen in (36).

- (36) *ođđa lága birra mánáidgárddiid* *várás*
 new law about [kindergartens.GEN.PL for]
 ‘the new law for kindergartens’

Thus, since all other nominal modifiers precede the noun in Saami, and postnominal modifiers follow the “postposition”, the postpositions are in fact case suffixes. Again, it is the word order of the modifiers that decides whether an element is an adposition or an affix.

5. Flags are prefixes when the noun must occur in initial position

Analogously and conversely, we find that in some languages, nominal modifiers always follow the noun, and that this leads to a situation in which preposed flags always occur on the noun, i.e. they are prefixes.

Consider Irish, which has postposed demonstratives (37a), adjectives (37b), adpossessioners (37c), and relative clauses (37d). There are no preposed modifiers.

- (37) Irish
- a. *na-daoine sin*
 DEM-people those
 ‘those people’

- b. *teach beag*
house small
'a small house'
- c. *bailte na hÉireann*
towns the Ireland.GEN
'the towns of Ireland'
- d. *post nua a thabharfaidh siad duit*
post new REL give.FUT.3PL they to.you
'a new job that they will give you'

As a result, Irish “prepositions” are in fact prefixes, and to make this point in a salient way, they are written as prefixes in (38a-c). In the conventional Irish spelling, they are written separately.

- (38) a. *i-bPáras* (i bPáras)
LOC-Paris
'in Paris'
- b. *ar-an-mbord* (ar an mbord)
LOC-DEF-table
'on the table'
- c. *do-mo-mháthair* (do mo mháthair)
BEN-1SG.POSS-mother
'for my mother'

It should be noted that the Irish definite article (*an-* in 38b) and possessor person forms (such as *mo-* in 38c) are prefixes, too: They cannot occur on their own, and they always occur on the noun. Thus, while the preposed flags *i-*, *ar-* and *do-* in (38) do not occur adjacent to the noun root, they still occur “on the noun” and are thus prefixes. (See also Ahlqvist (1974), and Hutchinson (2025) on the closely related Scottish Gaelic language.)

In Irish, even numerals (as in (39)) are prefixes, as they are bound forms that cannot occur on their own but must combine with an anaphoric noun (Stenson 2020: 199). This is illustrated in (40), which shows the anaphoric noun *cheann*.

- (39) *dhá-theach*
two-houses
'two houses'

- (40) *Tá trí theach agam, agus tá dhá cheann ag mo dheartháir.*
COP three houses at.me and COP two ones at my brother
'I have three houses, and my brother has two (lit. ones).'

In earlier Irish, adjectives could also precede the noun (Stifter 2010: 105), but in the modern language, this is only possible in a few residual compounds, e.g.

- (41) *sean-bhean* 'old woman'
sean-fhear 'old man'

Thus, when the order of adjectives (and other modifiers) became rigid, prepositions became case prefixes. Again, the change had nothing to do with the nature of the flags.

Similarly to Irish, Arabic has preposed flags that are prepositions in some varieties and prefixes in others. In Standard Arabic, they are prepositions, as in (42), but in Egyptian Arabic, they are prefixes, as in (43).⁷

(42) Standard Arabic

- a. مع خالي
maʕa xaal-ii
 with uncle-1SG
 ‘with my uncle’
- b. مع هذه الفتاة
maʕa haaðihī l-fataat-i
 with this.F DEF-girl-GEN
 ‘with this girl’

(43) Egyptian Arabic (Gary & Gamal-Eldin 1982: 63, 86)

- a. *maʕa-xaal-i*
 with-uncle-1SG
 ‘with my uncle’
- b. *maʕa-l-bint di*
 with-DEF-girl this
 ‘with this girl’

As demonstratives follow the noun in Egyptian Arabic, the preposed flags always occur on the noun, making them affixes. The only difference is a difference in element order. Again, the difference has nothing to do with the nature of the preposed flags.

Another Semitic language that is worth mentioning in this connection is Amharic, whose prepositions correspond to Proto-Semitic prefixes (assuming that all modifiers preceded the noun in Proto-Semitic, as in Egyptian Arabic; see Huehnergard 2019). Some examples of Amharic prepositions are given in (44a-c); it should be noted that in Amharic, they are written as prefixes, like the monoconsonantal flags in Arabic.

(44) Amharic

- a. ለወንድሙ
lä wändämm-u (läwändämmu)
 for brother-3SG
 ‘for his brother’ (Leslau 2000: 129)
- b. ለጥቂት ጊዜ
lä təqit gize (lätəqit gize)
 for little time
 ‘for a while’ (Leslau 2000: 129)

⁷ A reviewer rightly asks how we know that preposed demonstratives are not prefixes in Arabic, like Irish numerals. The answer is that they can occur on their own (Ryding 2005: 316; e.g. *haaðihī* ‘this (female) one’).

- c. በገበሬው እርሻ
bä gäbäre-w əršə (bägäbärew əršə)
 in farmer-DEF field
 ‘in the farmer’s field’ (Leslau 2000: 128)

So as Amharic developed more and more prenominal modifiers (under the influence of Cushitic; Greenberg 1980), as in (44b) and (44c), its earlier case prefixes inherited from Proto-Semitic must have become prepositions. The shapes or other properties of these elements did not change – the only thing that changed was the order of the nominal modifiers. (A caveat here is that the order of prenominal modifiers in Proto-Semitic cannot be reconstructed with certainty.)

6. The opposite direction: From case affix to adposition

Let us finally look at a well-known case of a seeming change from affix to clitic: English Genitive *'s*, and Swedish Genitive *-s*. These forms go back to a Proto-Germanic suffix *-s*, but they are generally regarded as clitic postpositions in modern English and modern Swedish, as they occur after some postnominal modifiers (e.g. *the king of Spain's decision*, where it occurs after a postnominal adpossessor). The apparent change from case suffix to enclitic (postposition) has often been discussed under the heading of “degrammaticalization” (e.g. Rosenbach 2008; Norde 2009).

According to Norde (2006: 205), older Swedish had a situation not unlike German, where the genitive marker *-s* occurs not only on the noun, but also on prenominal modifiers, as seen in (45).

- (45) older Swedish
en-s rik-s man-s hus
 one-GEN rich-GEN man-GEN house
 ‘a rich man’s house’

But in modern Swedish, the element *-s* only occurs on the last word of the adpossessor, which is typically the noun (as in 46a), but which can also be part of a postnominal modifier (as in 46b).

- (46) modern Swedish
 a. *en rik man-s hus*
 a rich man-GEN house
 ‘a rich man’s house’
 b. *en vän till mig-s företag*
 a friend of mine-GEN company
 ‘a friend of mine’s company’

What exactly was it that changed between older and modern Swedish? This is not clear to me. Were there postnominal modifiers such as *till mig* ‘of mine’ in older Swedish? If so, they did not contain the marker *-s*, so the order of this element would have changed. But maybe all such modifiers were prenominal in older Swedish, in which case the order of some kinds of modifiers changed. But again, it appears that the change had nothing to

do with the nature of the flag: It always sounded [s] and had genitive meaning, and whether it is regarded as affix or clitic depends on the word order. Thus, discussing it under the heading of “degrammaticalization” (Norde 2009) may not be very helpful.

7. A possible explanatory factor: The flexibility-length universal

So far in this paper, I have talked about the position of nominal modifiers, not about the position of flags from the perspective of their own properties. However, one might ask whether it is something about the properties of flags that makes them likely to occur only on nouns (and thus likely to become affixes). One candidate is the Flexibility-Length Universal in (47).

(47) The flexibility-length universal

In all languages, the longer a constituent is, the more likely it is to show flexible order.

This universal is not well-known and has not been discussed in the literature, as far as I am aware, but impressionistically, it seems to be broadly true. Clauses generally have more flexible order than nominal phrases, and phrases generally have more flexible order than individual words or morphs. And it seems to be true both for the external order and the internal order of constituents (for example, nominal phrases are internally less flexible than clauses, and they are also less flexible with respect to their own position).

Very short elements (such as particles or clitic pronouns) can occasionally occur in multiple positions, but most short elements occur in a rigid position, adjacent to some larger element – these short elements are often affixes. Why the flexibility-length universal should hold is unclear, but it could be related to perceptibility: If an element is longer and thus more salient, it can be easily perceived independently and does not need to cooccur with a longer element.

One might therefore suspect that shortness generally leads to rigid order, and that this explains why grammaticalization typically involves increasingly rigid order, just as it typically involves shortening. If case affixes were generally shorter than clitics, one might attribute their rigid order to their greater shortness, but it is doubtful that they are generally shorter (see Zingler & Rogers 2026 for a recent cross-linguistic study; however, their definition of “clitic” is apparently much narrower than (4b)).

However, clitics are often quite rigid in their order as well, and adpositions are almost always rigidly ordered on the periphery of their complement nominal. Thus, I am not optimistic that these considerations can truly help us explain changes from clitic to affix, including changes from adposition to case affix. But I mention it here as a speculative possibility, as I think that further research on the relation between length and positional flexibility would be valuable.

8. Does grammaticalization consist in loss of independence?

In this final section, let us look back at the key question that was asked at the beginning: Can we understand the change from clitic flag (or adposition) to affix flag (or case affix) in terms of grammaticalization, loss of independence, and increase in bondedness?

Intuitively, the nature of the change from adposition to affix is that a “separate word” loses its independence and becomes “part of a word”. But this intuition may be deceptive.

Whether “words” exist independently of our spelling-based intuitions is unclear (Haspelmath 2011a). Likewise, whether syntax and morphology are separate parts of grammars is unclear: The idea of a special “lexical component” has been losing ground in different corners of linguistics (e.g. Bruening 2018). It may be that the idea of a “lexicon” as a separate domain comes from the fact that linguists typically record information about languages in two different kinds of books, grammars and dictionaries (see also Haspelmath 2024).

Traditional ideas may sometimes have an undue influence on our thinking in typology, and the “suffixing preference” (Greenberg 1957; Bybee et al. 1990) may be an example of this. It is true that there could be discourse reasons for a general suffixing preference (Himmelmann 2014), but it may also be that the spelling is leading us astray. Consider again the Irish expression in (48).

- (48) *do-mo-mháthair*
to-my-mother
‘for my mother’

We saw that there are good reasons to treat *do-* and *mo-* as prefixes, but the ordinary spelling treats them as words: *do mo mháthair*. It could be that preposed grammatical elements tend not to be written as prefixes because Latin and Greek, the classical languages, have many more suffixes than prefixes, and European languages could have inherited an “orthographic suffixing preference” from these two highly influential languages. Specifically for the case of flags, it has sometimes been observed that there are far fewer case prefixes than case suffixes (Reh 1986), but it could be that this is due to the way affixes are traditionally identified (via the spelling, or under the strong indirect influence of the spelling).⁸

Thus, the widespread intuition that “independent elements” become “dependent parts” in grammaticalization may be based on traditional stereotypes, from which we should free ourselves. Instead of relying on complex traditional notions (such as “word” or “morphology”) that may not have any reality, it seems better to start with basic terms such as “morph”, “root”, “boundness”, and “selectivity”. Such concepts allow us to define the terms “affix” and “clitic” in a new way that is quite independent of spelling or phonology (Haspelmath 2021; 2023a), and that is more rigorous than traditional way of identifying these elements.

The idea that selectivity determines the distinction between case affixes and adpositions is in line with much of the literature on the contrast between affixes and clitics, but very different from the traditional view in the grammaticalization literature, according to which there is a continuous scale from content item to affix. Thus, Narrog & Heine (2021: 76) say:

“Gradual loss of morphosyntactic independence of the form undergoing grammaticalization typically proceeds along the scale described in [55]. The ultimate outcome can be loss of a morphological boundary between the grammaticalizing form and its host or some other form.

[55] free form (> clitic) > affix”

⁸ There is also another possibility: It could be that nominal modifiers (or at least some subclasses) have an inherent tendency to be preposed to nouns, so that there will be fewer languages (of the Irish type) with preposed flags and no preposed modifiers than languages with postposed flags and no postposed modifiers (of the Lezgian type). If this were the case, the relative rarity of case prefixes would be related to word order, not to “bondedness” or “lack of independence”. (However, for adjectives, Dryer (2018: 801) actually finds the opposite.)

Moreover, it has often been said that there are “degrees of grammaticalization”, as in the following quotation (Lehmann 1986: 11):

“We subsume the German and the Turkish case suffixes under a common term, and similarly the Latin and Abkhaz postpositions; we decide that German *zu* is an allative preposition, but English *to* is a dative preposition; ... For all these pairs of concepts, there is a common functional denominator and a difference in the degree of grammaticalization which we, intuitively or explicitly, use as a criterion in applying the terms to categories of various languages. Given that differences in the degree of grammaticalization can by definition only be gradual, these analytical concepts must be prototypical concepts. We all know certain focal instances of the concepts of case suffix and postposition, personal pronoun and personal affix and so forth, and we apply these terms to new phenomena depending on their similarity to our focal instances.”

However, if there is a continuous scale and “degrees” along this scale, we should be able to quantify this scale. This may be possible, but so far, linguists have applied the criteria in an intuitive and non-rigorous way (using metaphorical expressions such as “tightly bonded”). Moreover, linguists have typically applied multiple criteria, and the choice of criteria is often opportunistic (involving cherry-picking). For systematic quantitative research, we need to apply criteria that are applicable uniformly in all languages. (See Belyaev 2021 for some cross-linguistic work in this direction.)

The definitions of *affix* and *clitic* that I used here are applicable uniformly, but are they significant for our understanding of general diachronic tendencies? This is not clear, and even though the change from clitic to affix is largely unidirectional (with exceptions in Germanic, as seen in §6, and possibly in Amharic), I do not know how it should be explained in general. Moreover, grammaticalization should probably not be thought of as a change from “syntax to morphology”, or “morphologization”. Grammaticalization is probably best defined as in (49) (Haspelmath 2027b):

(49) grammaticalization

Grammaticalization is the change from a free form to a grammatical marker.

This definition does not include changes from clitic to affix, so such changes seem to be (somewhat or completely) independent of changes from free forms to grammatical markers. It is very similar to Boye’s (2023) definition (“grammaticalization as conventionalization of discursively secondary status”) because free forms are potentially primary in discourse by convention, while grammatical markers are necessarily in the background.

References

- Ahlqvist, Anders. 1974. Notes on “case” and word-boundaries. *Ériu* 25. 181–189.
- Baxter, Alan N. 1988. *A grammar of Kristang (Malacca Creole Portuguese)*. Canberra: Dept. of Linguistics, Research School of Pacific Studies, Australian National University. (<http://dx.doi.org/10.15144/PL-B95>)
- Belyaev, Oleg. 2021. Rethinking lexical integrity: Phrase-level and word-level case morphology. In Butt, Miriam & Findlay, Jamie Y. & Toivonen, Ida (eds.), *Proceedings of the LFG’21 Conference (On-Line)*, 5–25. Stanford: CSLI Publications. (<https://web.stanford.edu/group/cslipublications/cslipublications/LFG/LFG-2021/toc.shtml>)

- Bisang, Walter & Malchukov, Andrej & the Mainz grammaticalization project team. 2020. Universal and areal patterns in grammaticalization. In Bisang, Walter & Malchukov, Andrej (eds.), *Grammaticalization scenarios: Cross-linguistic variation and universal tendencies (Volume 1)*. Berlin: De Gruyter. (doi:10.1515/9783110563146)
- Boye, Kasper. 2023. Grammaticalization as conventionalization of discursively secondary status: Deconstructing the lexical–grammatical continuum. *Transactions of the Philological Society* 121(2). 270–292. (doi:10.1111/1467-968X.12265)
- Bruening, Benjamin. 2018. The lexicalist hypothesis: Both wrong and superfluous. *Language* 94(1). 1–42. (doi:10.1353/lan.2018.0000)
- Bybee, Joan L. & Pagliuca, William & Perkins, Revere D. 1990. On the asymmetries in the affixation of grammatical material. *Studies in typology and diachrony: Papers presented to Joseph H. Greenberg on his 75th birthday*, 1–42. Amsterdam: Benjamins.
- Bybee, Joan L. & Pagliuca, William & Perkins, Revere D. 1990. On the asymmetries in the affixation of grammatical material. *Studies in typology and diachrony: Papers presented to Joseph H. Greenberg on his 75th birthday*, 1–42. Amsterdam: Benjamins. (<https://www.torrossa.com/gs/resourceProxy?an=5016221&publisher=FZ4850#page=35>) (Accessed September 4, 2024.)
- Creissels, Denis. 2006. Suffixes casuels et postpositions en hongrois. *Bulletin de la Société de Linguistique de Paris* 101(1). 225–272.
- Dryer, Matthew S. 2018. On the order of demonstrative, numeral, adjective, and noun. *Language* 94(4). 798–833. (doi:10.1353/lan.2018.0054)
- Dryer, Matthew. 1989. Plural words. *Linguistics* 27. 865–896. (doi:10.1515/ling.1989.27.5.865)
- É. Kiss, Katalin & Hegedűs, Veronika (eds.). 2021. *Syntax of Hungarian: Postpositions and postpositional phrases*. Amsterdam: Amsterdam University Press. (doi:10.5117/9789463725910)
- Gajdarov, R. I. & Gjul'magomedov, A.G. & Mejlanova, U.A. & Talibov, B. B. 2009. *Sovremennyj lezginiskij jazyk [The modern Lezgian language]*. Maxačkala: IJaLI DNC RAN.
- Gary, Judith Olmsted & Gamal-Eldin, Saad. 1982. *Cairene Egyptian Colloquial Arabic* (Lingua Descriptive Studies, 6). Amsterdam: North Holland.
- Greenberg, Joseph H. 1957. Order of affixing: A study in general linguistics. *Essays in Linguistics*, 86–94. Chicago: University of Chicago Press.
- Greenberg, Joseph H. 1980. Circumfixes and typological change. In Traugott, Elizabeth Closs & Labrum, Rebecca & Shepherd, Susan (eds.), *Papers from the 4th International Conference on Historical Linguistics*, 233–241. Amsterdam: Benjamins.
- Haspelmath, Martin. 1993. *A grammar of Lezgian* (Mouton Grammar Library 9). Berlin: Mouton de Gruyter.
- Haspelmath, Martin. 2011a. The indeterminacy of word segmentation and the nature of morphology and syntax. *Folia Linguistica* 45(1). 31–80. (doi:10.1515/flin-2017-1005)
- Haspelmath, Martin. 2011b. The gradual coalescence into “words” in grammaticalization. In Narrog, Heiko & Heine, Bernd (eds.), *The Oxford handbook of grammaticalization*, 342–355. Oxford: Oxford University Press.
- Haspelmath, Martin. 2015. Defining vs. diagnosing linguistic categories: A case study of clitic phenomena. In Błaszczak, Joanna & Klimek-Jankowska, Dorota & Migdalski, Krzysztof (eds.), *How categorical are categories? New approaches to the old questions of noun, verb, and adjective*, 273–304. Berlin: De Gruyter Mouton.
- Haspelmath, Martin. 2021. Bound forms, welded forms, and affixes: Basic concepts for morphological comparison. *Voprosy Jazykoznanija* 2021(1). 7–28. (doi:10.31857/0373-658X.2021.1.7-28)
- Haspelmath, Martin. 2022. What is “phonological fusion”? A plea for clear concepts such as boundness or welding. *Diversity Linguistics Comment*. (doi:10.58079/nswu) (<https://dlc.hypotheses.org/2847>)
- Haspelmath, Martin. 2023a. Types of clitics in the world’s languages. *Linguistic Typology at the Crossroads* 3(2). 1–59. (doi:10.6092/issn.2785-0943/16057)

- Haspelmath, Martin. 2023b. Are clitics intermediate between lexemes and affixes in grammaticalization? *Diversity Linguistics Comment* (blogpost) (<https://dlc.hypotheses.org/3130>)
- Haspelmath, Martin. 2024. Four kinds of lexical items: Words, lexemes, inventorial items, and mental items. *Lexique: Revue en Sciences du Langage* (34). 71–95. (doi:10.54563/lexique.1737)
- Haspelmath, Martin. 2027a. Flags and flagging in general grammar: On case marking and adpositional marking (in preparation).
- Haspelmath, Martin. 2027b. Biverbal predicate constructions and related concepts: Serial verbs, converbs, auxiliaries, and grammaticalization (to appear).
- Himmelman, Nikolaus P. 2014. Asymmetries in the prosodic phrasing of function words: Another look at the suffixing preference. *Language* 90(4). 927–960.
- Himmelman, Nikolaus P. 2014. Asymmetries in the prosodic phrasing of function words: Another look at the suffixing preference. *Language* 90(4). 927–960.
- Hopper, Paul J. & Traugott, Elizabeth Closs. 2003. *Grammaticalization* (2nd edition). Cambridge: Cambridge University Press.
- Hualde, José Ignacio & Ortiz de Urbina, Jon (eds.). 2003. *A grammar of Basque* (Mouton Grammar Library 26). Berlin: Mouton de Gruyter.
- Huehnergard, John. 2019. Proto-Semitic. In Huehnergard, John & Pat-El, Na'ama (eds.), *The Semitic languages* (2nd edition), 49–79. London: Routledge.
- Hutchinson, John G. 2025. On prepositions and case in Scottish Gaelic. *Word Structure* 18(1–2). 1–40. (doi:10.3366/word.2025.0245)
- Julien, Marit. 2018. Head-initial postpositional phrases in North Sámi. In Bailey, Laura R. & Sheehan, Michelle (eds.), *Order and structure in syntax I*. Berlin: Language Science Press. (doi:10.5281/zenodo.1117686)
- Konvička, Martin. 2019. Grammaticalisation clines: A brief conceptual history. *History and Philosophy of the Language Sciences* 2019. (<https://hiphilangsci.net/2019/03/06/grammaticalisation-clines/>)
- Lehmann, Christian. 1986. Grammaticalization and linguistic typology. *General Linguistics* 26(1). 3–22.
- Lehmann, Christian. 2015. *Thoughts on grammaticalization*. Berlin: Language Science Press. (<http://langsci-press.org/catalog/book/88>) (Originally published in 1982)
- Leslau, Wolf. 2000. *Introductory grammar of Amharic*. Wiesbaden: Harrassowitz.
- Linzen, Tal & Kasyanenko, Sofya & Gouskova, Maria. 2013. Lexical and phonological variation in Russian prepositions. *Phonology* 30(3). 453–515. (doi:10.1017/S0952675713000225)
- Narrog, Heiko & Heine, Bernd. 2021. *Grammaticalization*. Oxford: Oxford University Press.
- Norde, Muriel. 2006. Demarcating degrammaticalization: the Swedish s-genitive revisited. *Nordic Journal of Linguistics* 29(2). 201–238.
- Norde, Muriel. 2009. *Degrammaticalization*. Oxford: Oxford University Press.
- Pescarini, Diego. 2021. *Romance object clitics: Microvariation and linguistic change*. Oxford: Oxford University Press.
- Reh, Mechthild. 1986. Where have all the case prefixes gone? *Afrikanistische Arbeitspapiere* 5. 121–134.
- Rosenbach, Anette. 2008. The English s-genitive: A case of degrammaticalization? In Fischer, Olga & Norde, Muriel & Perridon, Harry (eds.), *Up and down the cline: The nature of grammaticalization*, 73–96. Amsterdam: Benjamins.
- Ryding, Karin C. 2005. *A reference grammar of Modern Standard Arabic*. Cambridge: Cambridge University Press.
- Sneddon, James N. 1996. *Indonesian: A comprehensive grammar*. London: Routledge.
- Spencer, Andrew & Luís, Ana R. 2012. *Clitics*. Cambridge: Cambridge University Press.
- Spencer, Andrew. 2008. Does Hungarian have a case system? In Corbett, Greville & Noonan, Michael (eds.), *Case and grammatical relations: Studies in honor of Bernard Comrie*, 35–56. Amsterdam: Benjamins.

- Stenson, Nancy. 2020. *Modern Irish: A comprehensive grammar* (Routledge Comprehensive Grammars). London: Routledge.
- Stifter, David. 2010. Early Irish. In Ball, Martin John & Müller, Nicole (eds.), *The Celtic languages* (Routledge Language Family Series), 55-116. 2nd ed. New York: Routledge.
- Thuilier, Juliette. 2011. Case suffixes and postpositions in Hungarian. *Proceedings of the International Conference on Head-Driven Phrase Structure Grammar* 209–226. (doi:10.21248/hpsg.2011.12)
- Trommer, Jochen. 2008. “Case suffixes”, postpositions, and the phonological word in Hungarian. *Linguistics* 46(2). 403–437.
- Wiemer, Björn. 2014. Quo vadis grammaticalization theory? Why complex language change is like words. *Folia Linguistica* 48(2). 425–468. (doi:10.1515/flin.2014.015)
- Zingler, Tim & Rogers, Phillip. 2026. The size of clitics and affixes: A phonological approach to the grammaticalization cline. *Studies in Language* 50(2). 352–395. (doi:10.1075/sl.25003.zin)
- Zwicky, Arnold M. & Pullum, Geoffrey K. 1983. Cliticization vs. inflection: English *n't*. *Language* 59(3). 502–513.